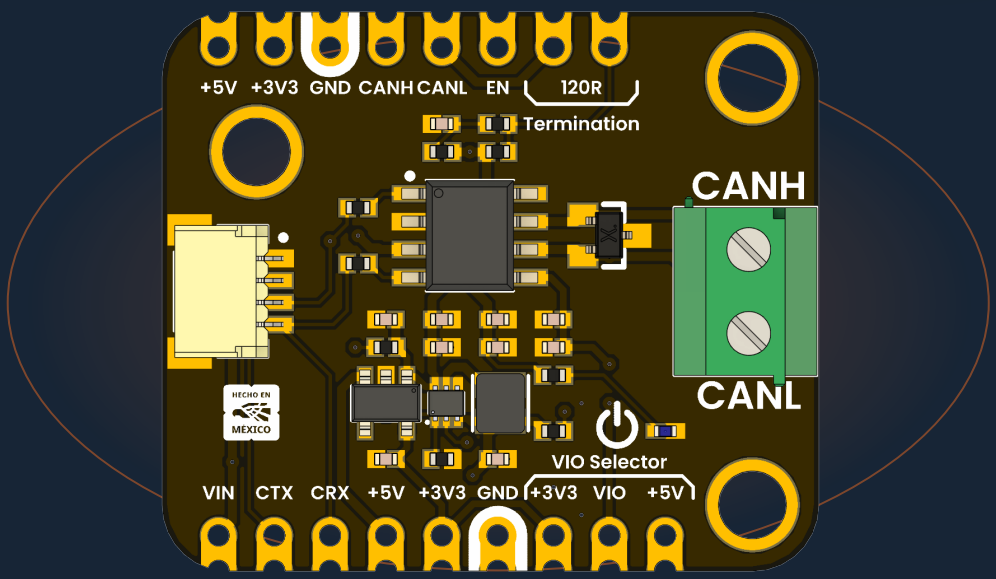
PRODUCT REFERENCE MANUAL

DEVLAB FACTOR

TCAN1051HVD CAN Transceiver Module

Mfr. Part #: UE0085

Version 1.0.0



Fault-protected high-speed CAN physical-layer interface with selectable 3.3 V or 5 V controller logic.

Classic CAN | CAN FD · 2 Mbps | 3.3 V / 5 V I/O | 120 Ω termination

DOCUMENT MAP

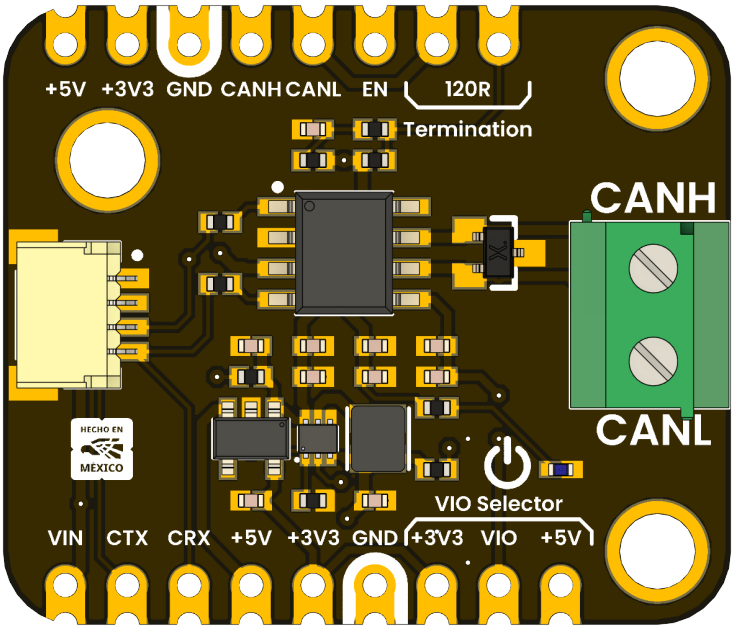
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Description

The **DevLab TCAN1051HVD CAN Transceiver Module** is a compact physical-layer interface for Classic CAN and CAN FD networks. It connects a microcontroller or stand-alone CAN controller to the differential CAN bus through the CTX (TXD), CRX (RXD), CANH, and CANL signals.

The module is built around the Texas Instruments TCAN1051HVD, a fault-protected high-speed CAN transceiver with a separate VIO supply. An onboard boost converter and 3.3 V regulator generate the transceiver and logic rails from VIN. The VIO selector allows direct use with either 3.3 V or 5 V controllers.



> This module implements the CAN physical layer only. The host must provide a CAN controller or a microcontroller with an integrated CAN/TWAI peripheral.

Applications

- Industrial automation and distributed control
- Robotics and mobile platforms
- Building and environmental control
- Automotive prototyping and diagnostics
- CANopen, DeviceNet, NMEA 2000, ISOBUS, and other CAN-based networks

Hardware Features

- TCAN1051HVD high-speed, fault-protected CAN transceiver
- Classic CAN and CAN FD operation up to 2 Mbps
- Selectable 3.3 V or 5 V logic interface (VIO)
- Onboard TPS61023 5 V boost converter and AP2112K 3.3 V regulator
- Switchable split 120 Ω termination (2 × 60 Ω with 4.7 nF center capacitor)
- PESD2CANFD24L-UX protection device on CANH and CANL
- 22 Ω series resistors on CTX and CRX
- 3.81 mm screw terminal for the CAN bus
- 4-pin, 1 mm-pitch connector for VIN, GND, CTX, and CRX
- Castellated headers for breadboard, carrier-board, or direct-wire integration
- Power indicator LED and exposed regulator enable signal

1 The Board

The DevLab TCAN1051HVD CAN Transceiver Module (UE0085) provides a physical-layer interface between a CAN-capable host controller and a differential CAN bus.

The design separates the controller-side logic interface from the CANH and CANL bus signals. Controller connections are available through a 4-pin JST-compatible connector and castellated contacts, while the CAN bus is accessible through a 3.81 mm screw terminal and dedicated board contacts.

An onboard 5 V boost converter, 3.3 V linear regulator, selectable logic voltage, and CAN bus protection circuitry support integration with 3.3 V and 5 V controllers.

1.1 Accessories

The UE0085 module is supplied with the following accessories:

Accessory	Quantity	Purpose
DevLab TCAN1051HVD CAN Transceiver Module (UE0085)	1	Main CAN physical-layer interface board
JST/Qwiic-compatible cable (AR4561)	1	Connects the module to the host controller through the 4-pin interface
Single-row male header, 1 × 9 pins	1	Provides solderable connections for board integration
Single-row male header, 1 × 8 pins	1	Provides additional solderable connections
Jumper bridge, 2.54 mm	2	Used for VIO logic-voltage selection (3.3 V or 5 V) and enabling the onboard 120 Ω CAN bus termination

The supplied headers and jumper bridges may require installation depending on the intended application.

Select only one VIO voltage level at a time. Enable the 120 Ω termination only when the module is installed at a physical end of the CAN bus.

1.2 Board Identification

Item	Value
Product	DevLab TCAN1051HVD CAN FD Transceiver Breakout
Brand / company	UNIT Electronics
Board ecosystem	DevLab
Product type	High-speed CAN / CAN FD physical-layer transceiver module
Main transceiver	Texas Instruments TCAN1051HVD
Power converter	TPS61023
Voltage regulator	AP2112K-3.3
Logic interface	Selectable 3.3 V / 5 V

Item	Value
CAN interface	CANH and CANL differential bus
Manufacturer Part Number (MPN)	UE0085
Available schematic	Revision 1
Product Reference	Version 1.0.0

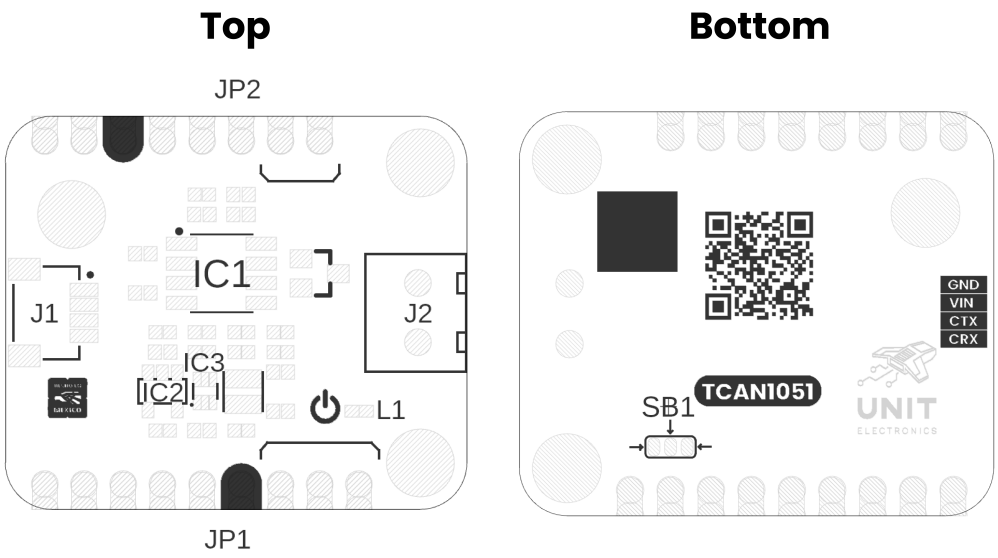
Board, schematic, pinout, and documentation revisions are controlled independently.

1.3 Identified Assemblies

Assembly	Function	Source status
TCAN1051HVD (IC1)	High-speed CAN physical-layer transceiver supporting Classic CAN and CAN FD up to 2 Mbps	Identified in the schematic and product reference
TPS61023 (U1)	Boost converter generating the regulated 5 V rail from VIN	Identified in the schematic
AP2112K-3.3 (U2)	Linear regulator generating the 3.3 V rail from the 5 V supply	Identified in the schematic
PESD2CANFD24L-UX (D1)	Transient protection for CANH and CANL	Identified in the schematic
Green LED (D2)	Indicates the presence of the regulated 5 V supply	Identified in the schematic
Controller connector (J1)	4-pin, 1 mm-pitch connection for VIN, GND, CTX, and CRX	Identified in the schematic and pinout
CAN bus terminal	2-position, 3.81 mm screw terminal for CANH and CANL	Identified in the schematic
Split termination network (R2, R3, C5)	Provides 120 Ω CAN bus termination using two 60 Ω resistors and a 4.7 nF center capacitor	Identified in the schematic
Termination selector (JP1)	Controls the connection of the onboard termination network	Identified in the schematic; assembly configuration requires verification
VIO selector (JP4)	Selects 3.3 V or 5 V transceiver logic levels	Identified in the schematic and pinout
Enable signal (EN)	Controls the TPS61023 boost converter	Identified in the schematic and pinout
Castellated contacts	Provide access to power, control, and CAN bus signals	Identified in the board documentation

The module implements the CAN physical layer only. A host microcontroller with an integrated CAN controller or an external CAN controller is required for communication.

1.4 Board Views



Views of Board Topology

Views of Board Topology

The top view identifies the main integrated circuits, controller and CAN bus connectors, power circuitry, and castellated contacts.

The bottom view shows the VIO solder selector, board identification markings, and controller signal references.

Reference	Description	Location
IC1	TCAN1051HVD CAN transceiver	Top
IC2	AP2112K-3.3 linear regulator	Top
IC3	TPS61023 boost converter	Top
J1	4-pin controller interface connector	Top
J2	2-position CAN bus screw terminal	Top
JP1	CAN bus termination selector	Top
JP2	Controller and power castellated contacts	Top
L1	Boost converter inductor	Top
SB1	VIO solder selector	Bottom
QR code	Board identification and product reference	Bottom
GND, VIN, CTX, CRX	Controller interface signal markings	Bottom

The board topology illustration identifies component placement and interface locations. Electrical connections and component values are defined by the released schematic.

Board artwork and schematic revisions must be verified independently before manufacturing or hardware modification.

1.5 Handling

Use standard electrostatic discharge (ESD) precautions when handling or installing the module.

Remove power before installing headers, changing jumper positions, or modifying the VIO selection.

Select only one VIO voltage level at a time. Connecting the 3.3 V and 5 V rails together may damage the module or the connected host.

Verify CANH and CANL polarity before connecting the module to an active CAN network.

Enable the onboard 120 Ω termination only when the module is installed at a physical end of the CAN bus. Intermediate nodes must not introduce additional termination resistance.

Do not connect external power directly to the regulated 3V3 or 5V outputs unless the application has been specifically validated for that configuration.

The module does not provide galvanic isolation between the host controller and the CAN bus.

2 Ratings and Electrical Characteristics

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VIN	Module input voltage	3.3	—	5.0	V
VCC	TCAN1051HVD bus supply	4.5	5.0	5.5	V
VIO	Selected I/O supply	2.8	3.3 / 5.0	5.5	V
VOUT(5V)	TPS61023 regulated output, nominal	—	5.0	—	V
VOUT(3V3)	AP2112K regulated output, nominal	—	3.3	—	V
IO(RXD)	Recommended CRX output-current range	−2	—	2	mA
TA	TCAN1051HVD operating free-air temperature	−55	—	125	°C
CAN FD	Supported data rate for TCAN1051HVD	—	—	2	Mbps

VIN represents the recommended module input voltage. VCC, VIO, I/O current, temperature, and CAN FD values are TCAN1051HVD specifications.

The 5 V and 3.3 V rails are generated by the onboard TPS61023 boost converter and AP2112K-3.3 linear regulator, respectively.

The complete assembled module has not been independently qualified over the full IC temperature range.

2.2 Selected Transceiver Characteristics

Unless otherwise noted, the following characteristics apply to the Texas Instruments TCAN1051HVD transceiver.

Symbol	Description	Min	Typ	Max	Unit
VIH	CTX high-level input voltage	0.7 × VIO	—	—	V
VIL	CTX low-level input voltage	—	—	0.3 × VIO	V
VOH	CRX high-level output voltage at −2 mA	0.8 × VIO	—	—	V
VOL	CRX low-level output voltage at 2 mA	—	—	0.2 × VIO	V
VO(DOM), CANH	Dominant CANH output voltage	2.75	—	4.5	V
VO(DOM), CANL	Dominant CANL output voltage	0.5	—	2.25	V
VOD(DOM)	Dominant differential output, 50–65 Ω load	1.5	—	3.0	V

Symbol	Description	Min	Typ	Max	Unit
VCM	Receiver common-mode range	−30	—	30	V
tPROP(LOOP)	Total loop delay	—	100 / 110	160 / 175	ns

2.3 Power Regulation Characteristics

The module integrates a TPS61023 boost converter and an AP2112K-3.3 low-dropout linear regulator.

TPS61023 — 5 V Boost Converter

Symbol	Description	Min	Typ	Max	Unit
VIN(IC)	Converter operating input voltage	0.5	—	5.5	V
VSTART	Converter startup input voltage	—	—	1.8	V
VOUT	Adjustable output voltage range	2.2	—	5.5	V
VOUT(MOD)	Configured module output voltage	—	5.0	—	V
fSW	Switching frequency, VIN > 1.5 V	—	1.0	—	MHz
ILIM	Valley switch current limit, typical	—	1	—	A

The TPS61023 generates the regulated 5 V rail from VIN.

The converter current-limit specification refers to the internal switching current and must not be interpreted as the available module output current.

The maximum external load on the 5 V rail has not been independently characterized.

AP2112K-3.3 — 3.3 V Linear Regulator

Symbol	Description	Min	Typ	Max	Unit
VIN(LDO)	Regulator input voltage, nominal	—	5.0	—	V
VOUT	Fixed regulated output voltage, nominal	—	3.3	—	V
IOUT	Rated regulator output-current capability	—	—	600	mA
VDROP	Dropout voltage	TBD	TBD	TBD	mV
TA	Regulator operating temperature	TBD	—	TBD	°C

The AP2112K-3.3 generates the module’s 3.3 V rail from the regulated 5 V supply.

The 600 mA rating represents the regulator IC capability, not the guaranteed external current available from the assembled module.

Available output current depends on regulator thermal dissipation, PCB layout, input voltage, and total module power consumption.

Dropout voltage and operating-temperature limits require confirmation against the exact AP2112K manufacturer datasheet.

2.4 Absolute Maximum Ratings

Absolute maximum ratings are stress limits, not normal operating conditions.

TCAN1051HVD — CAN Transceiver

Symbol	Description	Min	Max	Unit
VCC	TCAN1051HVD 5 V supply	−0.3	7	V
VIO	TCAN1051HVD I/O supply	−0.3	7	V
VBUS	CANH or CANL, H-suffix device	−70	70	V
VDIFF	CANH-to-CANL differential voltage, H-suffix device	−70	70	V
IO(RXD)	CRX output current	−8	8	mA
TJ	IC junction temperature	−55	150	°C
TSTG	IC storage temperature	−65	150	°C

The ±70 V bus-fault rating applies only to the specified CAN bus pins and must not be applied to VIN, CTX, CRX, EN, VIO, 3V3, or 5V.

Power Regulators

The TPS61023 and AP2112K-3.3 have independent absolute maximum ratings defined by their respective manufacturers.

Their absolute maximum input, output, and temperature ratings must not be interpreted as the operating limits of the complete UE0085 module.

2.5 Module Power and Thermal Considerations

The module uses the following power architecture:

- VIN supplies the TPS61023 boost converter.
- TPS61023 generates the regulated 5 V rail.
- AP2112K-3.3 generates the 3.3 V rail from 5 V.
- TCAN1051HVD operates from the regulated 5 V supply.
- VIO selects either the 3.3 V or 5 V logic rail.
- Both regulated rails are accessible through the module headers.

The AP2112K power dissipation can be estimated as:

$P_D \approx (V_{IN(LDO)} - V_{OUT}) \times I_{OUT}$

At a 5 V regulator input and 3.3 V output, the voltage difference is approximately 1.7 V.

The available external current on the 3.3 V and 5 V rails has not been independently qualified.

Do not exceed the thermal, electrical, or current limits of any individual component.

3 Functional Overview

3.1 Signal Path

CTX drives the TCAN1051HVD TXD input. A logic LOW requests a dominant bus state, and a logic HIGH requests a recessive state. The transceiver converts this logic signal into the CANH/CANL differential output.

CRX is driven by the TCAN1051HVD RXD output. CRX is LOW when the receiver detects a dominant bus state and HIGH when it detects a recessive state. The CRX output level follows the selected VIO rail.

The TCAN1051HVD includes dominant-timeout, thermal-shutdown, and undervoltage protection. The bus and logic pins present high impedance when the transceiver is unpowered.

IC1 pin S is unconnected in the revision 1 module schematic. The IC's internal pull-down selects normal transceiver mode by default; S is not available as a module control. A host's TWAI controller mode is independent of this IC pin.

3.2 Power Architecture

VIN feeds the TPS61023 boost converter, which generates the 5 V transceiver supply. The AP2112K then generates the 3.3 V rail. Both regulated rails are exposed on the castellated headers.

The VIO selector connects the transceiver I/O supply to either 3.3 V or 5 V:

Host logic	VIO selector
3.3 V MCU or CAN controller	3V3
5 V MCU or CAN controller	5V

Select exactly one side of the VIO solder selector. Bridging both sides shorts the 3.3 V and 5 V rails and can damage the module or host.

EN controls the TPS61023 regulator. The onboard bias enables the regulator for normal use; driving EN LOW disables the generated supply rails and transceiver.

3.3 Bus Termination

The onboard termination is a split 120 Ω network made from two 60 Ω resistors. Its midpoint is AC-coupled to ground by 4.7 nF to reduce common-mode noise.

Close the **120R Termination** selector only when this module is installed at one physical end of the CAN trunk. A correctly wired point-to-point bus normally has one 120 Ω terminator at each end (approximately 60 Ω measured between CANH and CANL with power removed). Leave the selector open at intermediate nodes.

When the selector is closed, the external termination remains connected even if IC1 is unpowered. The IC's high-impedance, unpowered bus behavior does not remove that board-level 120 Ω load.

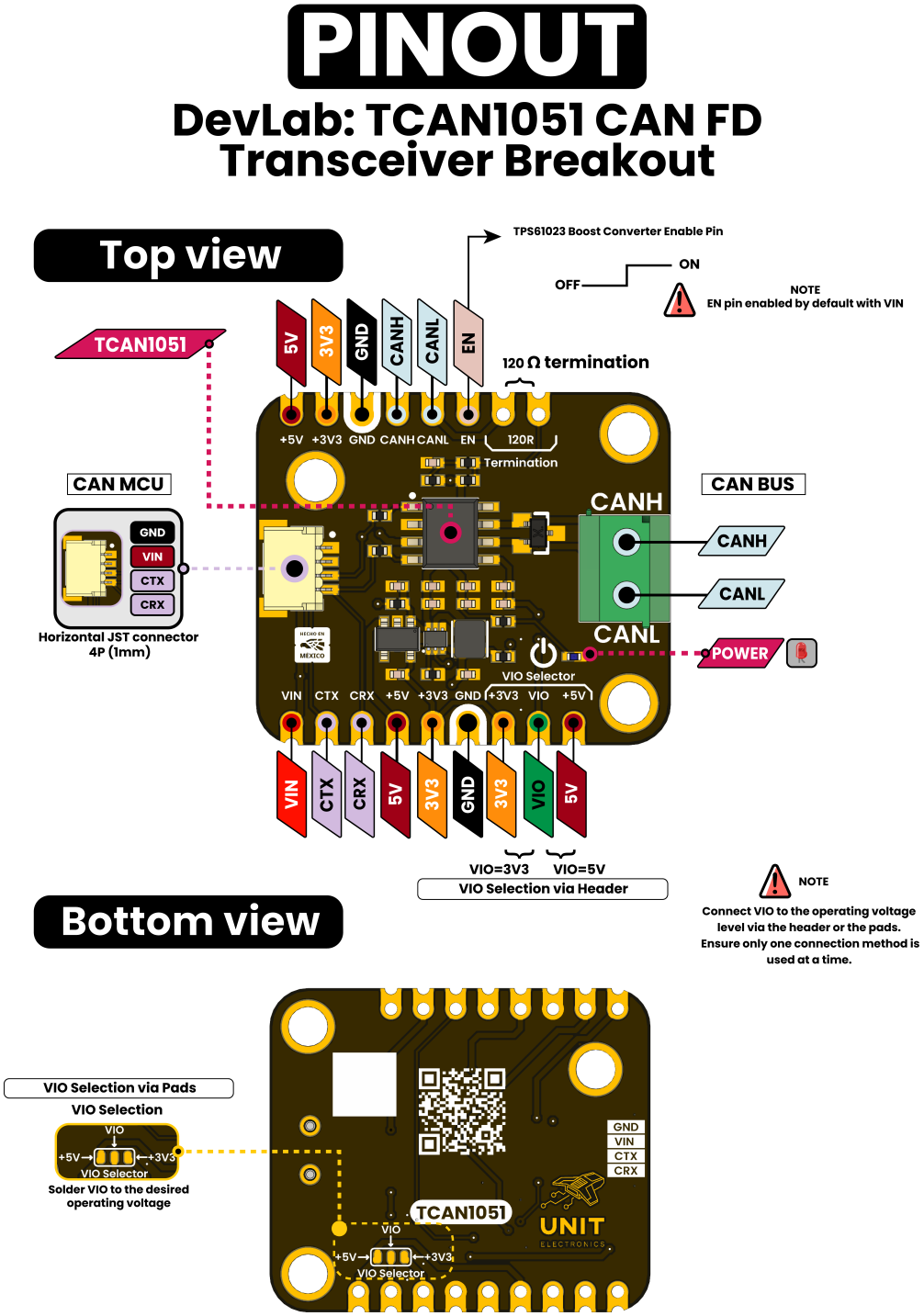
3.4 Bus Protection

CANH and CANL are protected by a PESD2CANFD24L-UX device. The TCAN1051HVD H-suffix transceiver provides a ±70 V bus-fault rating and a ±30 V receiver common-mode range. These protections improve robustness but do not replace correct grounding, cabling, termination, or system-level surge design.

4 Connectors and Pinouts

The UE0085 module provides controller-side logic connections, differential CAN bus terminals, and castellated contacts for integration with external hardware.

The following pinout identifies the available signals, power connections, and onboard configuration options.



4.1 Signal Reference

Label	Direction	Description
VIN	Power input	Module input supply, 3.3 V to 5.0 V
GND	Power	Common module and bus reference
CTX	Input	Controller transmit signal; LOW = dominant
CRX	Output	Controller receive signal; LOW = dominant
CANH	Bus I/O	CAN high bus line
CANL	Bus I/O	CAN low bus line
EN	Input	5 V boost-converter enable; LOW disables the generated supply rails
5V	Power output	Generated 5 V transceiver rail
3V3	Power output	Generated 3.3 V logic rail
VIO	Power	Selected transceiver logic supply (3.3 V or 5 V)

The CTX and CRX signals operate at the voltage level selected through VIO. The host controller must use compatible logic levels.

The module implements the CAN physical layer only. An external CAN controller or a microcontroller with an integrated CAN peripheral is required.

4.2 J1 Controller Connector

J1 is a 4-position, 1 mm-pitch connector providing power and communication signals to the host controller.

Use the bottom-side silkscreen to confirm the pin order before connecting a cable.

Signal	Connection
GND	Host ground
VIN	Host power supply
CTX	Host CAN TX output
CRX	Host CAN RX input

Verify the cable orientation and signal order before connecting the module to an external controller.

4.3 J2 CAN Bus Terminal

J2 is a 2-position, 3.81 mm screw terminal providing the differential CAN bus connections.

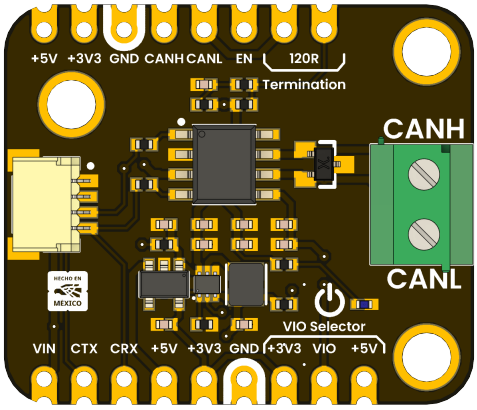
Position	Signal	Connection
Upper	CANH	CAN bus high line
Lower	CANL	CAN bus low line

Use twisted-pair wiring for CANH and CANL. Maintain correct bus polarity and minimize stub lengths to preserve signal integrity.

4.4 Castellated Headers

The castellated contacts provide access to controller signals, CAN bus connections, regulated power rails, and configuration signals.

These contacts support direct wiring, carrier-board integration, and prototyping applications.



The contact labels are printed on the board silkscreen. Refer to the pinout illustration for signal identification.

4.5 Configuration Jumpers

The module includes two jumper-selectable functions: VIO logic-voltage selection and CAN bus termination.

Configuration	Selection	Function
VIO	3V3	Sets the transceiver logic interface to 3.3 V
VIO	5V	Sets the transceiver logic interface to 5 V
CAN termination	Enabled	Connects the onboard 120 Ω termination network
CAN termination	Disabled	Disconnects the selectable termination network

VIO Selection

Select the VIO voltage according to the logic level of the connected host controller.

Only one VIO voltage selection must be active at a time. Connecting both the 3.3 V and 5 V rails together may damage the module or the host controller.

CAN Bus Termination

Enable the onboard 120 Ω termination only when the module is installed at a physical end of the CAN bus.

For intermediate nodes, leave the termination disabled.

A correctly terminated CAN bus normally uses one 120 Ω termination at each physical end of the main bus.

The termination network and jumper configuration must match the released hardware assembly revision.

5 Board Operation

This procedure brings up a two-node, 500 kbit/s Classic CAN bus. It uses one UE0085 module with a Pulsar ESP32-H2 transmitter and a second UE0085 module with an ESP32-P4 receiver. The module supplies the CAN physical layer; each ESP32 supplies the TWAI/CAN controller.

5.1 Required Hardware and Software

Item	Quantity	Purpose
UE0085 TCAN1051HVD module	2	One transceiver per controller
Pulsar ESP32-H2 and ESP32-P4 development boards	1 each	TWAI controllers
USB data cables and a computer	2	Uploads and serial monitoring
Twisted pair, hookup wires, common ground wire	As needed	Bus and host wiring
Regulated 3.3 V to 5.0 V supply for each module	1 per module	UE0085 VIN; a host 3.3 V rail is suitable only if it can supply the module
Multimeter	Recommended	Power and termination checks

Install Arduino IDE, an ESP32 board package supporting the selected boards, and the DevLab_TCAN1051HVD Arduino library. This library wraps the ESP32 TWAI peripheral; it does not add a CAN controller to a host without one. Confirm GPIO availability against the actual development-board revisions.

5.2 Hardware Connection and Configuration

Remove power before wiring or changing selectors. Each host needs its own UE0085 module. Connect J1 as follows; these GPIO assignments match the supplied example sketches.

UE0085 signal	Pulsar ESP32-H2 transmitter	ESP32-P4 receiver
VIN	Regulated 3.3 V to 5.0 V supply	Regulated 3.3 V to 5.0 V supply
GND	Host GND	Host GND
CTX	TWAI TX, GPIO12	TWAI TX, GPIO20
CRX	TWAI RX, GPIO22	TWAI RX, GPIO21

The controller-facing labels belong to the **UE0085 board**. The mounted TCAN1051HVD (IC1) uses different pin names:

IC1 pin	UE0085 connection	Function
1 TXD	CTX through R4 (22 Ω)	Logic input from the controller's CAN TX
2 GND	GND	Common reference
3 VCC	Onboard regulated 5V rail	Transceiver bus-side supply; not the module's VIN

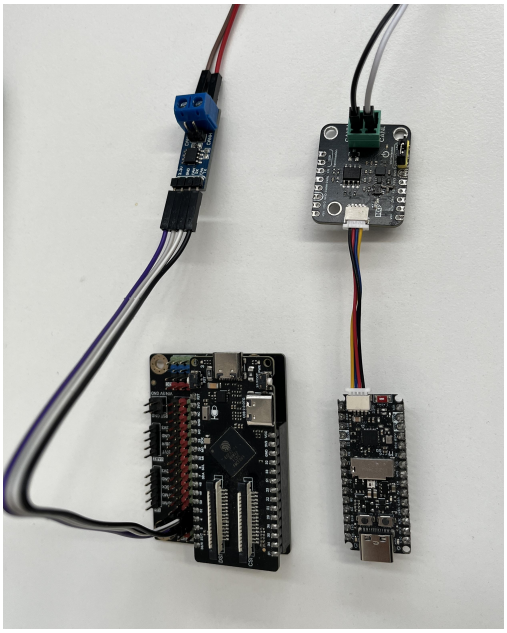
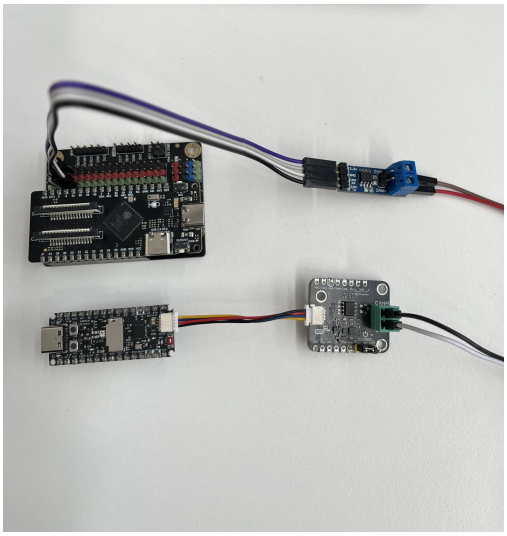
IC1 pin	UE0085 connection	Function
4 RXD	CRX through R5 (22 Ω)	Logic output to the controller's CAN RX
5 VIO	3V3/5V selector	Sets TXD/S input thresholds and RXD HIGH level
6 CANL, 7 CANH	J2 bus terminal, with onboard protection and selectable termination	Differential CAN bus
8 S	Unconnected on the revision 1 schematic	Internal pull-down selects normal transceiver mode by default; no external S control

VIN is the **module input**. The labeled 5V and 3V3 contacts are generated output rails, not substitutes for VIN. Set each module's VIO selector to **3V3** for these 3.3 V ESP32 GPIOs. Select exactly one VIO position; bridging 3V3 and 5V shorts the rails. Leave EN in its normal, onboard-enabled state. EN controls the boost converter; it does not select IC1's silent mode. For a 5 V host, select 5V only when its controller logic is 5 V compatible.

Connect the two J2 terminals **CANH to CANH** and **CANL to CANL** with a short twisted pair. Connect both module/host grounds as a common reference, even when the modules have separate supplies. Do not connect CANH or CANL to GPIOs. For this two-node bus, close the **120R Termination** selector on each module: the modules occupy the two physical ends. Do not add external terminators when both onboard networks are enabled. On a larger bus, terminate only its two physical ends and keep branch stubs short.

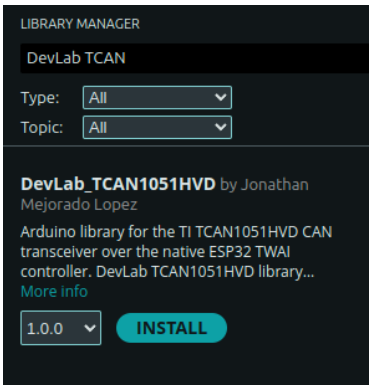
With all power removed, approximately **60 Ω** should measure between CANH and CANL on the assembled two-node bus. Approximately 120 Ω suggests a missing terminator; a much lower reading suggests extra termination or a short. After applying power, check that both green power LEDs illuminate and the 5V and 3V3 rails are near their nominal voltages. Remove power if a rail is shorted, the device heats unexpectedly, or the VIO selector bridges both rails.

The photographs below show an **illustrative mixed-transceiver bench setup**: the black UE0085 is connected to the ESP32-H2, while a different blue CAN transceiver is connected to the ESP32-P4. They show the physical arrangement, not the exact two-UE0085 wiring specified above. If using the blue module, follow its own pinout and supply requirements. In either setup, match CANH, CANL, ground reference, bit rate, and the two physical end terminations.

Connection overview	Closer view of the two CAN nodes
	

5.3 Arduino Library and Controller Setup

1. Install a compatible ESP32 board package in Arduino IDE. Select the exact ESP32-H2 or ESP32-P4 board and its serial port before each upload.
2. Install `DevLab_TCAN1051HVD` through Library Manager if listed. Otherwise download its repository as a ZIP and use **Sketch > Include Library > Add .ZIP Library**. Restart Arduino IDE if the header does not appear.
3. Open the two sketches in Section 5.4. The library also contains `CAN_Transmitter` and `CAN_Receiver` examples; this repository has `software/examples/basic_tx.ino` and `basic_rx.ino`.



Arduino Library Manager listing for DevLab_TCAN1051HVD; the displayed version may differ from the installed release

Both sketches explicitly use **500 kbit/s** and `TWAI_MODE_NORMAL`. Every node must use the same nominal bit rate. Normal mode requires another active node to acknowledge a transmitted frame. The example uses standard 11-bit identifier `0x100` and one data byte. The TCAN1051HVD supports the CAN FD physical layer up to 2 Mbps, but this ESP32-H2/P4 TWAI example uses **Classic CAN frames**, not CAN FD. `TWAI_MODE_NORMAL` selects the ESP32 controller mode; it does not drive IC1's unconnected S pin.

5.4 CAN Communication Example

Upload this transmitter sketch to the Pulsar ESP32-H2. It queues a frame containing an incrementing byte approximately once per second.

```
#include <DevLab_TCAN1051HVD.h>

DevLab_TCAN1051HVD can(GPIO_NUM_12, GPIO_NUM_22);
uint8_t counter = 0;

void setup() {
  Serial.begin(115200);
  if (!can.begin(TWAI_TIMING_CONFIG_500KBITS(), TWAI_MODE_NORMAL)) {
    Serial.println("CAN initialization failed");
    while (true) delay(1000);
  }
  Serial.println("Transmitter ready");
}

void loop() {
  uint8_t data[1] = {counter};
  if (can.send(0x100, data, 1, false, false, 1000)) {
    Serial.print("Frame queued, counter: ");
    Serial.println(counter);
    counter++;
  } else {
    Serial.println("CAN transmit queue failed");
  }
  delay(1000);
}
```

Upload this receiver sketch to the ESP32-P4. It prints the identifier and counter received from the bus.

```
#include <DevLab_TCAN1051HVD.h>

DevLab_TCAN1051HVD can(GPIO_NUM_20, GPIO_NUM_21);

void setup() {
  Serial.begin(115200);
  if (!can.begin(TWAI_TIMING_CONFIG_500KBITS(), TWAI_MODE_NORMAL)) {
    Serial.println("CAN initialization failed");
    while (true) delay(1000);
  }
  Serial.println("Receiver ready");
}

void loop() {
  TCAN1051_Frame frame;
  if (can.receive(frame, 100) && !frame.extended && !frame.rtr &&
      frame.id == 0x100 && frame.length == 1) {
    Serial.print("ID: 0x");
    Serial.print(frame.id, HEX);
    Serial.print("  Counter: ");
    Serial.println(frame.data[0]);
  }
}
```

The library's `send()` return value means the ESP32 TWAI driver accepted the frame into its transmit queue. By itself, it does **not** prove another node acknowledged or received the frame. The receiver output is the end-to-end communication check.

5.5 Upload and Communication Verification

1. With power off, recheck VIN, GND, CTX/CRX, CANH/CANL, both VIO selectors, and the two end terminators. Confirm about 60 Ω across CANH and CANL.
2. Power both boards and modules. Confirm both power LEDs illuminate.
3. Select the ESP32-P4 board and port, upload the receiver, and open its serial monitor at **115200 baud**. Wait for `Receiver ready`.
4. Select the Pulsar ESP32-H2 board and port, upload the transmitter, and open its serial monitor at **115200 baud**. Use separate IDE windows or serial terminals to watch both ports.
5. Confirm that the receiver repeatedly shows `ID: 0x100` and a counter advancing from 0, wrapping after 255. The transmitter should report queued frames at about one-second intervals.

Example receiver output (the library may also print an initialization line):

```
Receiver ready
ID: 0x100  Counter: 0
ID: 0x100  Counter: 1
ID: 0x100  Counter: 2
```

If only the transmitter reports queued frames, communication has not been verified. Check the receiver and bus using Section 5.6.

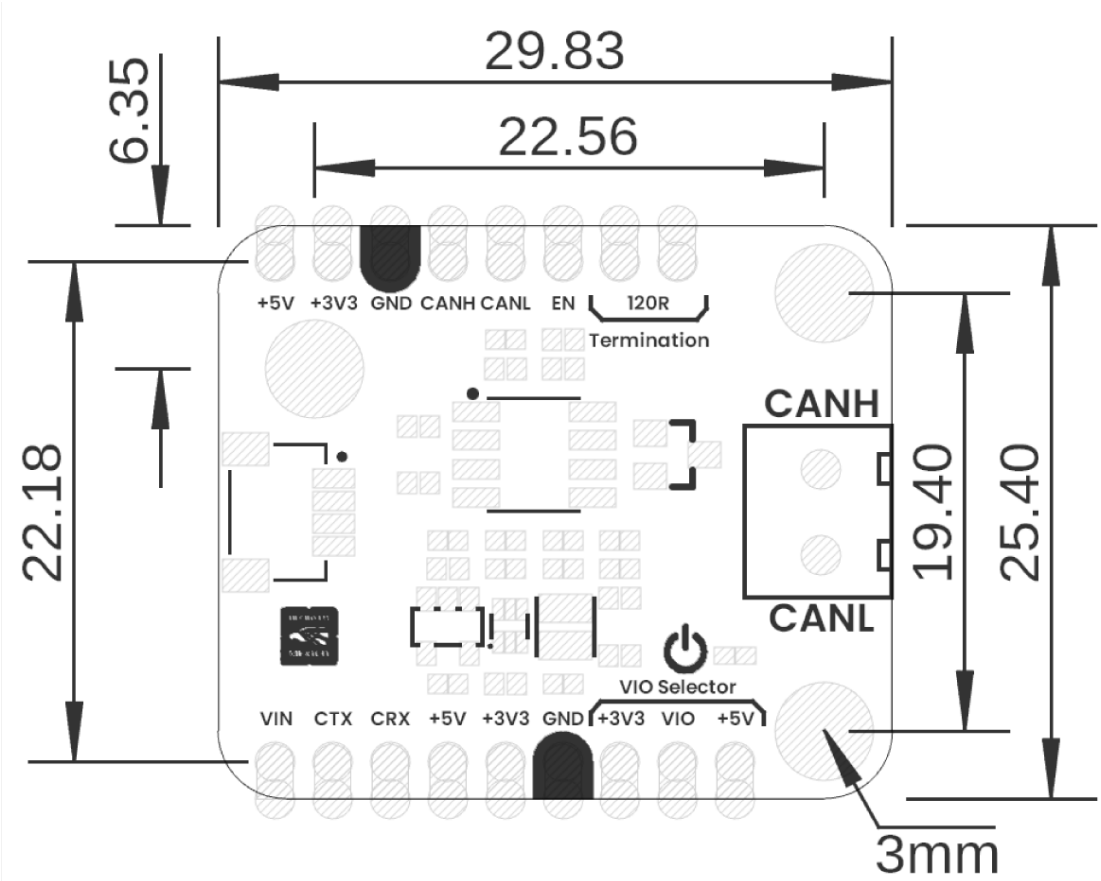
5.6 Troubleshooting

Symptom	Check
No power LED or incorrect rail voltage	VIN 3.3 V to 5.0 V, GND continuity, supply capacity, EN not held LOW, and no rail short.
Library header missing or compilation fails	Install `DevLab_TCAN1051HVD`, select a compatible ESP32 board package, and confirm `driver/twai.h` is available.
`CAN initialization failed`	Board selection, GPIO availability, TWAI support, and whether another sketch or driver has claimed TWAI.
Transmitter queues frames but receiver prints nothing	Receiver running, CTX/TX and CRX/RX assignments, CANH/CANL polarity, common ground, 500 kbit/s at both nodes, and 3V3 VIO.
Bus errors or repeated retransmission	Second active node in normal mode, one 120 Ω terminator at each end (about 60 Ω with power off), and matching bit rates. `send()` success alone is not an ACK indication.
Intermittent frames	Twisted pair, short stubs, sound terminals, ground reference, and bus length appropriate to the bit rate.
CTX stays LOW and the node stops transmitting	Check the TX GPIO configuration. IC1's TXD dominant timeout releases its bus driver; returning CTX HIGH clears that condition. Also inspect bus shorts and VIO selection.

For persistent faults, the library's `getStatus()` API can report TWAI state and bus-error counters. Correct wiring and termination before attempting bus-off recovery.

6 Mechanical Information

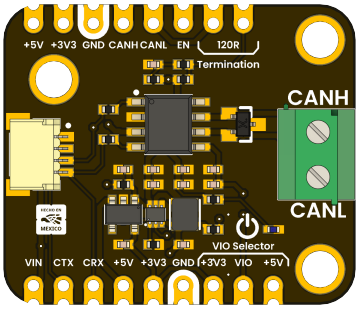
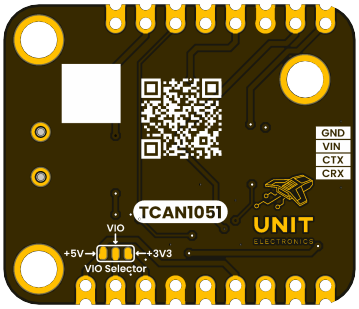
The nominal board outline is **29.83 mm × 25.40 mm**. The drawing below also identifies the 3 mm mounting holes and principal spacing dimensions. All values are in millimeters.



Board Dimensions in Millimeters

Use the released mechanical drawing or a physical sample when designing mating enclosures or production fixtures; the rendered image is not a 1:1 fabrication template.

6.1 Board Views

Top	Bottom
	

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Reference	Link
Product repository	UNIT-Electronics-MX/ unit_devlab_tcan1051hvd_can_transceiver_module
Arduino library	UNIT-Electronics-MX/unit_devlab_tcan1051hvd_library
Product Wiki	UNIT Electronics Wiki — UE0085
TCAN1051HV datasheet	Texas Instruments SLLSES8E
ISO 11898-2 overview	Refer to ISO 11898-2:2016 for the high-speed CAN physical layer

The released schematic, manufacturer datasheets, and hardware resources are available in the product repository under `hardware/` and `hardware/resources/datasheet/`.

The UNIT Electronics Wiki provides additional product information and integration references.

9 Appendix

9.1 Schematic

The complete revision 1 schematic is distributed with this product reference:

Open the UE0085 TCAN1051HVD module schematic

9.2 Design Notes

- R1 provides the switchable 120 Ω termination.
- R2 and R3 form the 2 × 60 Ω split termination.
- C5 (4.7 nF) AC-couples the split-termination midpoint to ground.
- R4 and R5 (22 Ω) are in series with CTX and CRX.
- D1 protects CANH and CANL.
- IC1 pin S is unconnected in the revision 1 schematic. The TCAN1051HVD's internal pull-down selects normal mode by default; the module does not expose S as a user control. For noisy or transient-prone systems, review this reliance on the IC's internal bias during hardware qualification.